

A person holds a concave shaving mirror 20 cm from their face. The mirror produces an upright image 1.5 times larger than the face. Calculate the focal length of the mirror. What would happen if they moved the mirror to 40 cm away?

A concave mirror is used to project an image of a candle flame onto a screen. The candle is 25 cm from the mirror and the screen is 75 cm from the mirror. Calculate the focal length and magnification. Is the image erect or inverted?

A dentist's mirror produces an upright image 4 times the actual size when held 3 cm from a tooth. What type of mirror is it, and what is its focal length?

A convex mirror has a radius of curvature of 40 cm. An object 6 cm tall is placed 20 cm in front of it. Find the position, nature, and size of the image formed.

A solar cooker uses a concave mirror of focal length 80 cm. Where should the cooking pot be placed to get maximum heat concentration? If sunlight falls parallel to the principal axis, where will the image of the sun be formed?

A convex mirror used as a rear-view mirror in a car has a focal length of -25 cm. If a car behind is actually 10 m away, calculate where its image appears and what is the magnification.

An object is placed in front of a concave mirror. The image formed is real, inverted, and twice the size of the object. If the image is formed 40 cm from the mirror, calculate:

Object distance
Focal length of the mirror
Radius of curvature

A concave mirror produces a virtual image 3 times larger than the object. If the object is 8 cm from the mirror, find the focal length and the distance of the image from the mirror.

Mathematical and Analytical Questions:

1. A concave mirror with radius of curvature 40 cm has an object placed at a distance that produces an image with magnification -2.5. Calculate the object distance, image distance, and determine whether the image is real or virtual.

Derive the relationship between the magnification of a spherical mirror and the object distance when the image distance is kept constant at 30 cm.

A convex mirror produces images that are always $\frac{1}{4}$ the size of the object. What is the focal length of this mirror, and at what distance should objects be placed to achieve this magnification?

For a concave mirror with focal length f , prove that the minimum distance between a real object and its real image is $4f$, and determine the magnification at this minimum distance.

A spherical mirror produces an image that is 3 times larger than the object when the object is placed 15 cm from the mirror. If the same object is moved to 45 cm from the mirror, calculate the new magnification and image characteristics.

Q1: Two plane mirrors are placed at 60° to each other. An object is placed between them. How many images will be formed?

- **Formula:** $n = (360^\circ/\theta) - 1$
- **Solution:** $n = (360^\circ/60^\circ) - 1 = 6 - 1 = 5$ images

Q2: Two mirrors at 45° have an object between them. Find total images and draw the positions.

- **Formula:** $n = (360^\circ/45^\circ) - 1 = 8 - 1 = 7$ images

2. Distance and Position Problems:

Q3: Two parallel mirrors are 20 cm apart. An object is 8 cm from the left mirror. Find distances of first 4 images from both mirrors.

- **Method:** Images alternate between mirrors
- Left mirror images: 8 cm, 32 cm, 52 cm...
- Right mirror images: 12 cm, 28 cm, 48 cm...

Q4: A person 1.6 m tall stands 2 m from a plane mirror. Where does their image appear and what is its height?

- **Answer:** Image appears 2 m behind mirror, height = 1.6 m (same as object)

3. Minimum Mirror Size Problems:

Q5: A person 1.8 m tall wants to see their full image. What is the minimum height of mirror needed and where should it be placed?

- **Formula:** Minimum mirror height = **Object height** $\div 2$
- **Answer:** Mirror height = $1.8 \div 2 = 0.9$ m
- **Position:** Bottom of mirror should be at height = Eye level - (Height above eye level $\div 2$)

4. Rotation Problems:

Q6: A plane mirror rotates 30° . By how much does the reflected ray rotate?

- **Formula:** Reflected ray rotation = **$2 \times$ Mirror rotation**
- **Answer:** Reflected ray rotates by $2 \times 30^\circ = 60^\circ$

Q7: A mirror rotates at 5 rpm. Find the angular speed of reflected ray.

- **Formula:** Angular speed of reflected ray = **$2 \times$ Angular speed of mirror**
- **Answer:** $2 \times 5 = 10$ rpm

5. Complex Geometry Problems:

Q8: Three mirrors form an equilateral triangle. A candle is placed at the center. How many images are formed?

- **Method:** Each pair of mirrors at 60° creates 5 images
- **Total calculation:** Use combination of mirror pairs

Q9: Two mirrors at 90° have an object 5 cm from each mirror. Find the distance between the two closest images.

- **Formula:** For perpendicular mirrors, closest images are $5\sqrt{2}$ cm apart
- **Method:** Use Pythagorean theorem

6. Practical Applications:

Q10: A periscope uses two parallel mirrors 30 cm apart. If an object is 50 cm from the top mirror, where does the final image appear?

- **Answer:** Final image appears 50 cm below the bottom mirror

Q1: A convex lens of focal length 20 cm forms an image at 60 cm. Find the object distance and magnification. What type of image is formed?

- **Given:** $f = +20$ cm, $v = +60$ cm
- **Find u:** $1/20 = 1/60 - 1/u \rightarrow 1/u = 1/60 - 1/20 = -2/60 \rightarrow u = -30$ cm
- **Magnification:** $m = v/u = 60/(-30) = -2$ (inverted, real, magnified)

Q2: An object 5 cm tall produces a virtual image 15 cm tall through a lens. If the object is 10 cm from the lens, find the lens type and focal length.

- **Given:** $h = 5$ cm, $h' = 15$ cm, $u = -10$ cm
- **Magnification:** $m = h'/h = 15/5 = 3$ (virtual, erect)
- **Since $m = v/u$:** $3 = v/(-10) \rightarrow v = -30$ cm
- **Focal length:** $1/f = 1/v - 1/u = 1/(-30) - 1/(-10) = 2/30 \rightarrow f = +15$ cm (convex lens)

2. Power and Combination Problems:

Q3: Two lenses of power +5D and -3D are placed in contact. Find the combined focal length and power.

- **Combined power:** $P = P_1 + P_2 = 5 + (-3) = +2D$
- **Combined focal length:** $f = 1/P = 1/2 = 0.5$ m = 50 cm

Q4: A lens of focal length 25 cm is cut into two equal parts along its principal axis. What is the focal length of each part?

- **Answer:** Each part has the same focal length = 25 cm (focal length doesn't change)

Q9: A myopic person can see clearly up to 40 cm. What power lens is needed to see distant objects clearly?

- **For distant objects:** $u = \infty$, v should be -40 cm
- **$1/f = 1/(-40) - 1/\infty = -1/40$**
- **Power needed:** $P = -1/0.4 = -2.5D$ (concave lens)

Q10: A hypermetropic person's near point is 50 cm instead of 25 cm. What power lens corrects this?

- **To read at 25 cm:** $u = -25 \text{ cm}$, $v = -50 \text{ cm}$
- **$1/f = 1/(-50) - 1/(-25) = 1/50$**
- **Power needed:** $P = +1/0.5 = +2\text{D}$ (convex lens)